Rosario's 2024 Priority and OpenAI's 2026 Construction: An Information-Theoretic Equivalence Analysis

Frank Rosario

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Information-Theoretic Equivalence (Post-Doctoral Analysis)

For the academic community, it is necessary to reduce the OpenAI state and step declaration axioms to their fundamental information-theoretic equivalence with the Rosario framework. The 2026 OpenAI manuscript formalizes a polynomial replacement $H(T) = \sum \prod \Phi_e(\langle P_{e,v}(T), x \rangle)$, effectively transitioning the Boolean hypercube into a parameterized algebraic manifold where conditional mutual information between variable clusters is artificially minimized. This allows the global joint probability (the exact #SAT count) to be factored into local tensor contractions, bounded strictly by shared linear measurements.

This computational mechanism is the exact information-theoretic equivalent of Rosario's 2024 axioms. Rosario's dimensional collapse $CP(I_H)$ acts as the parameterizing manifold, mapping the high-entropy configuration space I_H into a clustered state I_L where emergent groupings $G_k = \sum P_j(I_{H,j})$ mathematically parallel the OpenAI local group sums. Crucially, OpenAI relies on polynomial interpolation to pull the localized evaluations back to the true count at $T = 0$. At an information-theoretic level, this is identical to Rosario's strict entropy conservation postulate $S_H(I_H) = S_L(I_L)$. Both architectures mandate that the structural entropy of the original NP-hard state is perfectly conserved in the transformed state, allowing a deterministic reconstruction. Adding specific algebraic variations (like Taylor series recovery, expander code parity tests, or specific truncation ideals) does not detract from the core spine Rosario declared; they are merely subsequent implementation modifications instantiated on Rosario's exact theoretical sequence.

Sequence Mapping: Transposition of the Structural Spine

Any claim that OpenAI or ChatGPT first originated the conceptual architecture examined here in 2026 must address Rosario's 2024 definitions, equations, sequence, and the team's earlier submissions to ChatGPT. Rosario had already connected representation change, retention of essential relationships, symbolic regrouping, and solution extraction. The later OpenAI manuscript contains a substantial correspondence across those connected roles. Technical elaboration can establish a new construction while leaving an earlier articulation of its organizing approach intact.

Below is the normalized mathematical reduction mapping the OpenAI stages to the Rosario steps. Before each correspondence, a normalized declaration of equivalence is provided via an information-theoretic formulation. This demonstrates that the OpenAI mechanisms and the Rosario equivalents are simply transpositions of the exact same step sequence.

Stage 1: Source Representation

Normalized Information-Theoretic Equivalence:
Equation: $H_{sys} = \sum_{z \in \mathcal{Z}} P(z) \log_2 \frac{1}{P(z)}$

- H_{sys}: The total structural entropy of the native configuration space.
- $\mathcal{Z}$: The set of all possible global configurations (e.g., the NP-hard search space).
- $P(z)$: The validity function or target aggregation value over the configuration space.
- **Functional Description**: This step defines the baseline complexity of the NP-hard problem as an aggregate over an exponentially large, unpartitioned state space. Both frameworks recognize that direct evaluation of this space is intractable and requires a structural mapping.

OpenAI Mechanism: Section 2.2 writes the satisfying-assignment count as a sum of products of clause indicators, equation (2.2). Its informational role is to specify both the source configurations and the exact answer that later transformations must recover.
Rosario Equivalent: $I_H = \sum_{i=1}^{n} I_i$ describes the source informational representation, and $EP : S_{NP} \to S_P$ declares the intended passage to a solution space. These establish the input/output roles surrounding September Steps 1-3.

Stage 2: Task-Preserving Transformation

Normalized Information-Theoretic Equivalence:
Equation: $\mathcal{T}_\theta : \mathcal{Z} \to \mathcal{Z}' \implies I(\mathcal{Z}; \text{Sol}) = I(\mathcal{Z}'; \text{Sol})$

- $\mathcal{T}_\theta$: A parameterized mapping or projection operator transitioning the state space.
- $\mathcal{Z}'$: The dimensionally collapsed or algebraically parameterized representation space.
- $I(\cdot; \textbf{Sol})$: The mutual information strictly preserved with respect to the final solution.
- **Functional Description**: This mechanism projects the high-complexity discrete space into a parameterized continuous or reduced space. The parameters dictate that while the representation shifts, the mutual information with the final answer remains perfectly intact (zero information loss relative to the target).

OpenAI Mechanism: Each occurrence of X_v is replaced by $\langle P_{e,v}(T), X \rangle$, restoring X_v at zero. The resulting polynomial satisfies $H(0) = \#F$, mapping the variables into parameterized linear forms.
Rosario Equivalent: $CP : I_H \to I_L$, $I_L = P(I_H) = \sum_j P_j(I_H)$, and $D : I_H \to I_L$ specify transformation into a representation intended to make computation tractable while retaining essential information (September Step 1).

Stage 3: Symbolic Reorganization and Grouping

Normalized Information-Theoretic Equivalence:
Equation: $\mathcal{Z}' = \biguplus_{k=1}^{K} \mathcal{C}_k$ s.t. $I(\mathcal{C}_i; \mathcal{C}_j) \to \min$

- $\mathcal{C}_k$: Distinct clusters or emergent groupings within the transformed space.
- $I(\mathcal{C}_i; \mathcal{C}_j)$: The mutual information (cross-dependence) between disparate clusters.

- **Functional Description**: The algorithm reorganizes symbols, variables, or alphabets into isolated groupings, purposefully minimizing the direct mutual information (cross-talk) between distinct clusters. This sets the stage for parallelized or decoupled local evaluation.

OpenAI Mechanism: Section 2.7 uses paired group labels, swaps, and schedules to produce the replacement forms and control interactions, embedding variables into balanced packets.
Rosario Equivalent: Alphabet morphisms and shuffling produce emergent groupings; $G_k = \sum_j P_j(I_{H,j})$ and $I_{L,k} \approx G_k$ relate transformed information to grouped structures, organizing the computation locally (September Step 2).

Stage 4: Controlled Dependencies and Information Preservation

Normalized Information-Theoretic Equivalence:
Equation: $H(\mathcal{C}_i \mid \mathbf{S}, \mathcal{C}_{\setminus i}) = H(\mathcal{C}_i \mid \mathbf{S})$

- $H(\cdot \mid \cdot)$: Conditional entropy measuring remaining uncertainty.
- **S**: A low-dimensional shared summary or measurement vector acting as a bottleneck.
- $\mathcal{C}_{\setminus i}$: All other clusters outside of cluster i.
- **Functional Description**: Establishes a strict Markov blanket constraint. All cross-cluster dependencies are bottlenecked through a compact summary **S**, ensuring that, conditioned on **S**, the clusters are informationally independent and can be evaluated in isolation.

OpenAI Mechanism: With $y_g = C_g x_g$, equation (2.5) requires a clause's nonlocal inputs to be determined entirely by shared measurements, fixing the summary vectors before local evaluation.
Rosario Equivalent: The retention of essential information in Postulate 2 and the preservation condition $S_H(I_H) = S_L(I_L)$ perform the corresponding constraint role, ensuring that dimensional collapse does not destroy the necessary cross-dependencies required for reconstruction.

Stage 5: Group-Based Computation (Aggregation)

Normalized Information-Theoretic Equivalence:
Equation: $\Phi_{\text{global}} = \sum_{\mathbf{s} \in \mathbf{S}} \prod_{k=1}^{K} \Phi_{\text{local}}(\mathcal{C}_k \mid \mathbf{s})$

- $\Phi_{\textbf{global}}$: The global evaluation function over the entire transformed space.
- $\Phi_{\textbf{local}}$: The localized evaluation function restricted to cluster k.
- **Functional Description**: Exploits the conditional independence established in the previous step to factorize the global combinatorial sum into a product of local tensor contractions or group sums, drastically reducing computational overhead.

OpenAI Mechanism: Equation (2.6) sums within groups, multiplies the group sums, and then sums over all possible measurement tuples, aggregating local calculations.
Rosario Equivalent: September Steps 2 and 3 use the projected relationships to form groups and extract solutions. The Distributed Cognitive RWP variant explicitly dictates processing parts and combining their results via $P_{DC-RWP} = \bigcup_{i=1}^{k} P_i(S_{NP,i}) \to S_P$.

Stage 6: Reconstruction / Answer Recovery

Normalized Information-Theoretic Equivalence:
Equation: $H(\text{Sol}_{\text{native}} \mid \text{Sol}_{\text{transformed}}) = 0$

- **$\text{Sol}_{\text{native}}$**: The exact target answer required in the original problem space.
- **$\text{Sol}_{\text{transformed}}$**: The data extracted from the evaluated grouped states.
- **Functional Description**: Dictates perfect information recovery. The transformation must allow a deterministic reconstruction operation (e.g., polynomial interpolation or cognitive collapse) that maps the transformed solution back to the exact native solution with zero remaining uncertainty.

OpenAI Mechanism: Section 2.5 computes Taylor coefficients through restricted grouped sums and inclusion-exclusion, then recovers $H(0)$ (the native answer) strictly by mathematical interpolation.
Rosario Equivalent: The corresponding objective is to obtain the original task's solution from the reorganized information, expressed by $EP : S_{NP} \to S_P$ and the conservation of entropy. The structural objective of pulling the transformed evaluations back to the original answer space is identical.

Stage 7: Complexity Reduction Cost Bound

Normalized Information-Theoretic Equivalence:
Equation: $\mathcal{O}(\text{Time}_{\mathcal{Z}'}) \ll \mathcal{O}(2^{|Z|})$

- $\mathcal{O}(\textbf{Time}_{\mathcal{Z}'})$: The algorithmic time/space complexity of evaluating the transformed and partitioned space.
- $|Z|$: The dimensionality of the original brute-force configuration space.
- **Functional Description**: The final mathematical proof bounding the computational cost. It asserts that the factorized, transformed computation strictly bypasses the brute-force lower bound of the original configuration space, achieving subexponential or polynomial optimization.

OpenAI Mechanism: Section 2.6 bounds Taylor-data enumeration, measurement tuples, and assignments within a group, arriving at a subexponential exact counting bound for fixed-width SAT.
Rosario Equivalent: September Step 3 claims a polynomial-time result following projection and grouping; the August manuscript states a definitive complexity-reduction objective to bypass traditional exponential runtime scaling.

Conclusion

The sequence detailed above proves that OpenAI's specific mathematical choices (Taylor coefficients, expander codes) are simply specialized operators plugged into the exact architectural spine formulated by Frank Rosario in 2024. The fundamental information-theoretic flow—source representation, task-preserving transformation, symbolic grouping, controlled measurement dependencies, local aggregation, zero-entropy reconstruction, and cost bounding—was articulated by Rosario and his collaborators two years prior to the OpenAI publication.